Name: _____

Calculations I=Q-t and V=E-Q

Date:

Time:

Total marks available: 17

Total marks achieved: _____

Questions

Q1.

There is a current of 2.5 mA in a circuit.

Calculate the amount of charge that moves past a point in the circuit in 80 seconds.

State the unit of charge.

(3)

charge = ... unit .....

(Total for question = 3 marks)

Q2.

The circuit in Figure 1 is switched on.

A charge of 1.2 C leaves the battery in a time of 4.0 s.

Calculate the current in the circuit.

Use the equation

$$\text{current} = \frac{\text{charge}}{\text{time}}$$

current = ... A

(Total for question = 2 marks)

Q3.

The voltage (potential difference) across a length of wire is 1.5 V.

A charge of 0.042 C flows through the wire.

Calculate the energy transferred.

Use the equation

$E = Q \times V$

(2)

E = ... J

(Total for question = 2 marks)

Q4.

(i) There is a current of 0.46 A in a lamp.

Calculate the total charge that flows through the lamp in 30 seconds.

Use the equation

charge = current × time in seconds

(2)

charge = ... C

(ii) The voltage across the lamp is 6.0 V.

The current in the lamp is 0.46 A.

Calculate the energy transferred to the lamp in one minute.

Use the equation

energy transferred = current × voltage × time in seconds

(2)

energy transferred = ... J

(Total for question = 4 marks)

Q5.

A torch has a battery and a bulb.

The current in its circuit is 0.08 A.

Calculate the amount of charge passing a point in this circuit in 2 minutes.

(3)

charge = ... coulombs

Q6.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 5 shows a charger for a car battery.

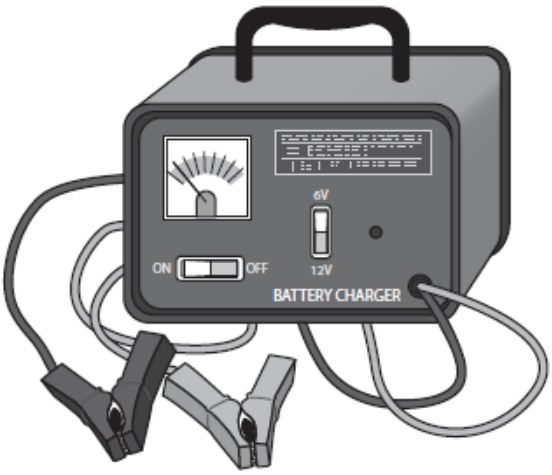

Figure 5

(i) The meter on the battery charger shows the current supplied to a battery.

The meter on the battery charger is

(1)

☐ **A** an ammeter

☐ **B** an ohmmeter

☐ **C** a voltmeter

☐ **D** a wattmeter

(ii) The battery charger supplies a steady current of 2.5 A to the battery.

Calculate the charge flowing to the battery in 8 minutes.

Use the equation

charge = current × time

(2)

charge = ... C

(Total for question = 3 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
	substitution into Q = I x t (1) (Q =) 2.5 (x 10^{-3}) x 80 evaluation (1) (Q =) 0.2 unit (1) coulombs OR C OR As	 allow 0.2 to any other power of ten for 1 mark award first two marks for correct answer without working independent mark allow c award 3 marks for 200 mC OR 200 mAs award 2 marks for 200 C OR 200 As	**(3)** **AO2**

Q2.

Question	Answer	Additional guidance	Mark
	substitution (1) (current =) $\frac{1.2}{4(.0)}$ evaluation (1) (current =) 0.3(0) (A)	 award full marks for the correct answer without working	**(2)** **AO2.2**

Q3.

Question number	Answer	Additional guidance	Mark
	substitution (1) (E =) 0.042 x 1.5 evaluation (1) (E =) 0.063 (J) (1)	 6.3×10^{-2} award 2 marks for the correct answer without working accept 0.063 to any other power of 10 for 1 mark	**(2)** **AO2**

Q4.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) (charge) = 0.46 x 30 evaluation (1) (charge) = 14 (C)	 any number that rounds to 14 e.g. 13.8 award full marks for the correct answer without working	**(2)** **AO2**

Question number	Answer	Additional guidance	Mark
(ii)	substitution (1) (energy transferred) = 0.46 x 6.0 x 60 evaluation (1) (energy transferred) = 170 (J)	 allow (energy transferred) = 0.46 x 6.0 x 1 or (energy transferred) = 0.46 x 6.0 x 30 any number that rounds to 170 e.g. 165.6 or 166 allow answers that round to 2.8 or 83 e.g. 2.76 or 82.8 for 1 mark only award full marks for the correct answer without working	**(2)** **AO2**

Q5.

Question Number	Answer	Acceptable answers	Mark
	Conversion to correct units: 120 seen anywhere (1) Substitution: 0.08 x 120 (1) Evaluation: 9.6 (C) (1) accept 10 C	Allow full marks for correct answer with no working seen. 0.08 x 2 gains 1 mark for sub of their time into correct eq'n 0.16 (C) gains 2 marks (only mistake is not converting time to seconds) accept any power of 10 error for 2 marks e.g. 960 (C)	**(3)**

Q6.

Question number	**Answer**	**Additional guidance**	**Mark**
i	A an ammeter B, C and D are incorrect because they do not show current		**(1)**

Question number	**Answer**	**Additional guidance**	**Mark**
ii	substitution (1) (Q) = 2.5 x 8 (x60) evaluation (1) 1200 (C)	 allow 20 (C) for 1 mark award full marks for the correct answer without working	**(2)**